

Shape-Constrained Additive Models

GAM.jl Contributors

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Introduction

This vignette demonstrates shape-constrained additive models using the R **scam** package, fitting the same simulated data as the Julia vignette for comparison.

Setup

```
library(scam)
```

This is scam 1.2-19.

```
library(mgcv)
```

Loading required package: nlme

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

Example 1: Monotone increasing (dose-response)

Load data

True function: $f(x) = 3(1 - e^{-5x})$

```
dat <- read.csv("../data.csv")
x <- dat$x
y <- dat$y
n <- nrow(dat)
f_true <- 3.0 * (1.0 - exp(-5.0 * x))
```

Fit unconstrained GAM vs SCAM

```
m_gam <- gam(y ~ s(x, k = 15, bs = "cr"), data = dat)
m_scam <- scam(y ~ s(x, k = 15, bs = "mpi"), data = dat)
```

Compare fitted values

```
yhat_gam <- predict(m_gam)
yhat_scam <- predict(m_scam)

rmse_gam <- sqrt(mean((yhat_gam - f_true)^2))
rmse_scam <- sqrt(mean((yhat_scam - f_true)^2))

cat("RMSE (unconstrained GAM):", round(rmse_gam, 4), "\n")
```

RMSE (unconstrained GAM): 0.0596

```
cat("RMSE (SCAM, monotone increasing):", round(rmse_scam, 4), "\n")
```

RMSE (SCAM, monotone increasing): 0.0501

Verify monotonicity

```
diffs_scam <- diff(yhat_scam)
diffs_gam <- diff(yhat_gam)

cat("Min successive difference (SCAM):", round(min(diffs_scam), 6), "\n")
```

Min successive difference (SCAM): 1e-06

```
cat("All non-decreasing (SCAM):", all(diffs_scam >= -1e-10), "\n")
```

All non-decreasing (SCAM): TRUE

```
cat("Min successive difference (GAM):", round(min(diffs_gam), 6), "\n")
```

Min successive difference (GAM): 1e-06

```
cat("All non-decreasing (GAM):", all(diffs_gam >= -1e-10), "\n")
```

All non-decreasing (GAM): TRUE

Example 2: Convex function

Simulate data

True function: $f(x) = 2x^2$

```
dat_cx <- read.csv("../data_cx.csv")
x_cx <- dat_cx$x
y_cx <- dat_cx$y

dat2 <- data.frame(y = y_cx, x = x_cx)
f_true2 <- 2 * x_cx^2
```

Fit with convexity constraint

```
m_cx <- scam(y ~ s(x, k = 15, bs = "cx"), data = dat2)

yhat_cx <- predict(m_cx)
rmse_cx <- sqrt(mean((yhat_cx - f_true2)^2))
cat("RMSE (convex SCAM):", round(rmse_cx, 4), "\n")
```

RMSE (convex SCAM): 6.7673

Verify convexity

```
first_diffs <- diff(yhat_cx)
second_diffs <- diff(first_diffs)
cat("Min second difference:", round(min(second_diffs), 6), "\n")
```

Min second difference: -0.061069

```
cat("All convex:", all(second_diffs >= -1e-10), "\n")
```

All convex: FALSE

Example 3: Monotone increasing and concave

Simulate data

True function: $f(x) = 3\sqrt{x}$

```
dat_micv <- read.csv("../data_micv.csv")
x_micv <- dat_micv$x
y_micv <- dat_micv$y

dat3 <- data.frame(y = y_micv, x = x_micv)
f_true3 <- 3 * sqrt(x_micv)
```

Fit with monotone increasing + concave constraint

```
m_micv <- scam(y ~ s(x, k = 15, bs = "micv"), data = dat3)

yhat_micv <- predict(m_micv)
rmse_micv <- sqrt(mean((yhat_micv - f_true3)^2))
cat("RMSE (monotone increasing + concave):", round(rmse_micv, 4), "\n")
```

RMSE (monotone increasing + concave): 1.5225

Verify constraints

```
first_diffs_micv <- diff(yhat_micv)
second_diffs_micv <- diff(first_diffs_micv)
cat("Min first difference (monotonicity):", round(min(first_diffs_micv), 6), "\n")
```

Min first difference (monotonicity): 0

```
cat("Max second difference (concavity):", round(max(second_diffs_micv), 6), "\n")
```

Max second difference (concavity): 0.032084

```
cat("Monotone increasing:", all(first_diffs_micv >= -1e-10), "\n")
```

Monotone increasing: TRUE

```
cat("Concave:", all(second_diffs_micv <= 1e-10), "\n")
```

Concave: FALSE

SCAM model summaries

```
summary(m_scam)
```

Family: gaussian

Link function: identity

Formula:

$y \sim s(x, k = 15, bs = "mpi")$

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.41623	0.02038	118.6	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(x)	3.408	4.201	337.9	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.8768 Deviance explained = 87.9%

GCV score = 0.084902 Scale est. = 0.083031 n = 200

```
summary(m_cx)
```

Family: gaussian

Link function: identity

Formula:

$y \sim s(x, k = 15, bs = "cx")$

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.74232	0.02041	85.37	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Approximate significance of smooth terms:
      edf Ref.df      F p-value
s(x) 1.857  2.189 289.6  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Rank: 14/15

R-sq.(adj) =  0.7606   Deviance explained = 76.3%
GCV score = 0.08451   Scale est. = 0.083303   n = 200

```

```
summary(m_micv)
```

```

Family: gaussian
Link function: identity

Formula:
y ~ s(x, k = 15, bs = "micv")

Parametric coefficients:
      Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.591117    0.007384   80.06  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
      edf Ref.df      F p-value
s(x) 3.393  3.714 160.9  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Rank: 14/15

R-sq.(adj) =  0.7491   Deviance explained = 75.3%
GCV score = 0.011149   Scale est. = 0.010904   n = 200

BFGS termination condition:
1.99322e-05

```

Comparison table

Feature	R <code>scam</code>	Julia <code>GAM.jl scam</code>
Monotone increasing	<code>bs="mpi"</code>	<code>bs=:mpi</code>
Monotone decreasing	<code>bs="mpd"</code>	<code>bs=:mpd</code>
Convex	<code>bs="cx"</code>	<code>bs=:cx</code>
Concave	<code>bs="cv"</code>	<code>bs=:cv</code>
Mono. inc. + convex	<code>bs="micx"</code>	<code>bs=:micx</code>
Mono. inc. + concave	<code>bs="micv"</code>	<code>bs=:micv</code>
Mono. dec. + convex	<code>bs="mdcx"</code>	<code>bs=:mdcx</code>
Mono. dec. + concave	<code>bs="mdcv"</code>	<code>bs=:mdcv</code>